

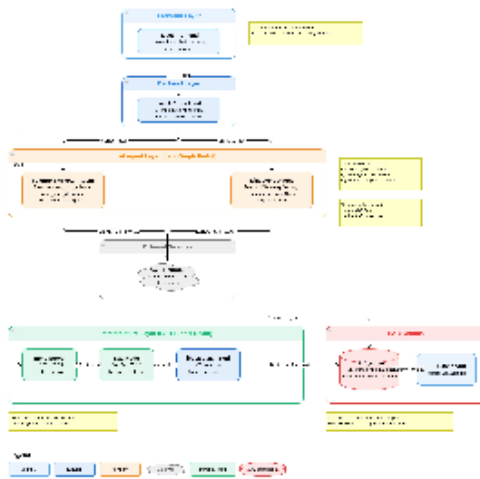
STC_EAI_DataReconciliationconfig

Scope

EAI Agentic Data Reconciliation is a end-to-end automated platform for generating mapping Excel files from NMS (Network Management System) JSON data and triggering discovery jobs on Ericsson Adaptive Inventory (EAI). It covers automated JSON schema analysis, natural language rule interpretation, .docx specification parsing, mapping Excel generation using a reference template, human-in-the-loop review with conversational refinement, automated file upload to Kubernetes pods via double-hop SSH, and discovery job triggering via EAI REST APIs. The platform uses two LangGraph ReAct agents powered by Claude Sonnet 4.5 on AWS Bedrock to autonomously reason, act, and observe through the mapping and discovery workflow.

Architecture

Architecture Diagram:



- Frontend — React + Vite

- Single-page application with a 5-step workflow wizard.
- Communicates with backend via REST API.
- Key views:
 - Input Configuration (JSON + Spec + NL Rules upload)
 - Feature Extractor Agent (pipeline visualization)
 - Human Review & Refine (mapping table with chat-based revision)
 - Upload to Pod (K8s upload status)
 - Discovery Agent (job trigger and EAI dashboard link).

- Backend — FastAPI (Python)

- Central orchestrator exposing REST endpoints for file upload, feature extraction, mapping preview, revision, pod upload, discovery job trigger, and workflow reset.
- Manages application state including uploaded file paths and NL rules. Serves the generated Excel for preview and download.

- **AI Agent Layer — 2 LangGraph ReAct Agents**

- Feature Extractor Agent — analyzes NMS JSON schema, parses .docx mapping specification, interprets natural language rules, and generates a mapping Excel file using a reference template that preserves lookup sheets.
- Discovery Agent — fetches the existing Discovery Job Definition (DJD) key from EAI and submits a ResourceSynchronizationRequest to trigger the reconciliation process.
- Both agents use Amazon Bedrock (Claude Sonnet 4.5, us-east-1) as the LLM provider with temperature=0 for deterministic outputs.

- **Security Scanner — 5 Tools**

- `analyze_json_schema` — reads the NMS JSON file, discovers equipment types, field names, counts, and sample values for each type.
- `parse_spec_document` — reads a .docx specification document and extracts all text content including tables, providing the mapping rules to the agent.
- `generate_mapping_excel` — takes structured JSON blocks from the agent, copies the reference Excel template (preserving all lookup sheets), and replaces the HuaweiShelf sheet with the generated mapping rules.
- `fetch_djd_key` — calls EAI GET API to retrieve the Discovery Job Definition key by name.
- `submit_discovery_job` — calls EAI POST API to submit a ResourceSynchronizationRequest using the DJD key.

- **Storage**

- Local filesystem — uploaded files (uploads/), generated Excel (output/), reference template (2_ATK_huawei_jsonmapping_01.xlsx).
- Prompt files — feature_extractor_prompt.txt and discovery_prompt.txt define agent workflows.

- **External Dependencies**

- AWS Bedrock — LLM provider for both agents.

- EAI REST APIs — discovery configuration, DJD lookup, job submission.
- K8s Cluster — hosts the jboss-eap74 pod with the ATK discovery adapter.
- SFTP/SSH Jump Server — access path to the K8s cluster.

Demo Mode & Latency Optimizations

• Mode Selector

The application now presents a mode selection screen on launch, allowing users to choose between:-

1. **Demo Mode** — All inputs (NMS JSON, mapping spec document, and NL rules) are pre-loaded from bundled sample data. The user walks through the same full workflow but without needing to locate or prepare any input files. Designed for presentations where the demonstrator may not have domain-specific knowledge of the input data.
2. **Live Mode** — Standard workflow where users upload their own files, write custom NL rules, and run the agents against real data. The selected mode is displayed as a badge in the application topbar and persists throughout the session. Resetting the workflow returns the user to the mode selection screen.

• Latency Optimization

Step	Live Mode	Demo Mode
Feature Extractor	Real agent (~30–40s)	Cached output (~6s with animations)
Revision	Full agent re-run (~25–40s)	Direct Excel edit (~3s)
Upload to Pod	Real SSH (~5s)	Real SSH (~5s)
Discovery Agent	Real API (~5s)	Real API (~5s)

1. **Feature Extractor in Demo Mode:** The agent output is served from a cached Excel file (cached_mapping_output.xlsx) that was pre-generated using the bundled sample inputs. The frontend animations simulate the three-step pipeline (Analyze Schema Parse Spec Generate Excel) at 2-second intervals for a realistic visual experience.

2. **Revision in Demo Mode:** For the four sample revision queries available in the chat interface, the backend applies the requested changes directly to the Excel file using openpyxl — the Excel is genuinely modified and the preview reflects real changes. A 3-second delay simulates agent processing time. Supported sample revisions:

- Change \$refId prefix (e.g., Shelf_ SH_)
- Add a new attribute with a fixed value (e.g., vendor = Huawei)
- Remove an attribute row (e.g., serialNumber)
- Change the object action rule (e.g., createOrUpdate lookup)

Any query not matching a known pattern falls through to the real agent for processing.

• Backend Initialization

Two new endpoints manage mode lifecycle. This ensures mode isolation — switching between demo and live mode does not leak cached results or stale state:

1. **POST /api/init-demo** — Copies bundled sample files (HuaweiSourceFile.json, OCI Site Excel Generation Spec.docx) into the working directory, sets NL rules in application state, enables the demo mode flag, and pre-loads the cached Excel output.
2. **POST /api/init-live** — Disables the demo mode flag, clears application state and any previously generated outputs, ensuring a clean environment for real data processing.

• Bundled Demo Assets

The following files are shipped with the application container for demo mode:

1. HuaweiSourceFile.json — Sample NMS inventory with 13 equipment items (1 SHELF type)
2. OCI Site Excel Generation Spec.docx — Mapping specification document defining column structure and processing rules
3. output/cached_mapping_output.xlsx — Pre-generated mapping Excel for the oci/shelf target type
4. Pre-configured NL rules for oci/shelf mapping (parser class, attribute definitions, parent associations, dynamic attributes)

UC's

• UC1 — Upload NMS JSON and mapping specification

A user uploads their NMS JSON inventory file and a .docx document containing the mapping specification. They also provide natural language rules describing which target types to generate and how to map attributes.

• UC2 — Generate mapping Excel via Feature Extractor Agent

The ReAct agent analyzes the JSON schema to discover equipment types and fields, reads the .docx spec to understand column definitions and processing rules, then generates a mapping Excel with the correct structure. The reference template is used as a base to preserve lookup sheets (LookupShelf, LookupStatus, etc.) required by the adapter.

• UC3 — Human review and conversational refinement

The generated mapping is displayed as a table grouped by target blocks. The user can review each block and either approve it or ask the agent to revise specific attributes via a chat interface. Revisions re-run the feature extractor agent with the updated instructions, regenerating the Excel.

• UC4 — Upload files to the K8s pod

The approved mapping Excel and source JSON are uploaded to the EAI pod via a double-hop SSH connection (jump server K8s node kubectl cp). Files are renamed to match what the ATK adapter expects (2_ATK_huawei_jsonmapping_01.xlsx and HuaweiSourceFile.json).

• UC5 — Trigger discovery job via Discovery Agent

The Discovery Agent fetches the DJD key for the pre-configured Test_Huawei definition and submits a

ResourceSynchronizationRequest. The EAI adapter automatically processes the mapping Excel, reads the source JSON, and triggers a BulkLoadRequest to reconcile the data into the inventory.

- **UC6 — View reconciliation results**

After the job completes, the user opens the EAI Dashboard directly from the application to view the reconciliation results, job status, and any errors.

- **UC7 — Reset and re-run workflow**

The user can reset the entire workflow to start fresh with new inputs. This clears all uploaded files, generated outputs, and state.

Demo Link

[AIFirst- Syncup-20260505_114109-Meeting Recording.mp4](#)

Presentation link

[Agentic EAI Flow 140426.pptx](#)